Bus Reservation System

A console-based C++ application for efficient bus ticket management

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Project Overview

Core Purpose

Manage bus ticket bookings efficiently through an automated console-based system

Key Functions

Reserve seats, view available buses, cancel reservations, and check booking details

Technology

Built using C++ with file handling for persistent data storage

This system reduces manual work and ensures accurate record-keeping for passenger reservations.

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Project Objectives

01

User-Friendly Interface

Provide an intuitive menu-driven interface for booking bus tickets with clear navigation

02

Record Management

Maintain comprehensive records of passengers, reservations, and seat availability

03

Error Reduction

Eliminate common errors found in manual booking systems through automated validation

04

C++ Concepts

Demonstrate practical application of classes, objects, file handling, and functions

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System Features

Bus Display

View all available buses with details including routes and departure times

Seat Reservation

Reserve specific seats with real-time availability checking

Ticket Cancellation

Cancel existing reservations and free up seats for other passengers

Booking Details

View complete reservation information including passenger details

Data Persistence

Store all data using file handling for permanent record-keeping

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System Modules

The project is organized into 6 functional modules for modular programming

Add Bus

Register new buses to the system with route and capacity details

Display Bus Details

Show comprehensive information about all available buses

View Reserved Seats

Display all currently reserved seats with passenger information

Search Bus

Find specific buses based on route, departure time, or other criteria

Ticket Cancellation

Cancel existing bookings and update seat availability

Seat Reservation

Reserve available seats and generate booking confirmations

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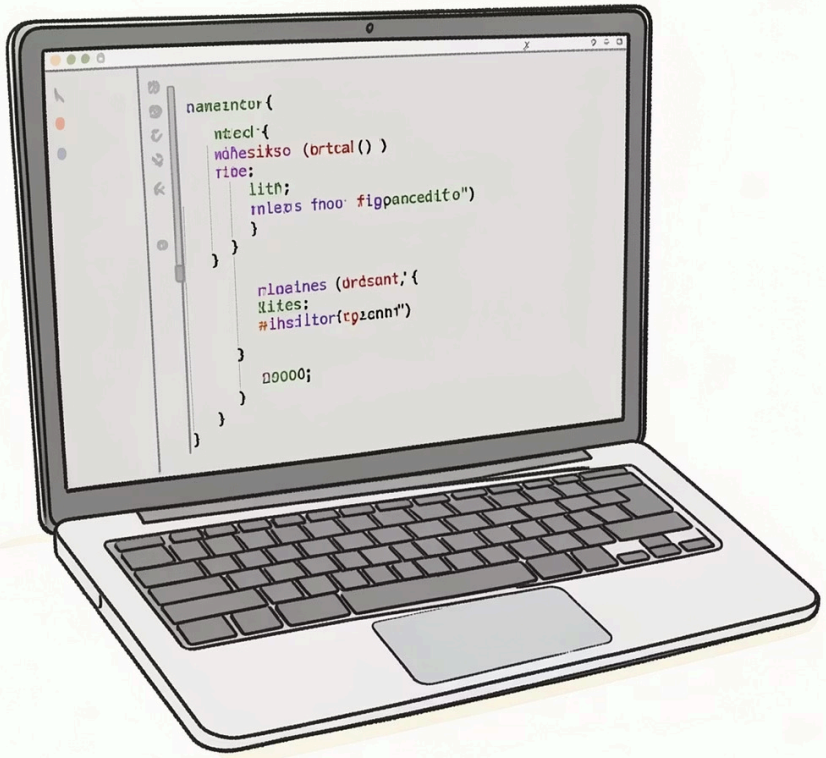
System Requirements

Hardware Requirements

- Computer or Laptop with minimum 4GB RAM
- Standard keyboard and display
- At least 100MB of free storage space

Software Requirements

- C++ Compiler (Dev-C++ or Code::Blocks)
- Windows or Linux operating system
- No additional libraries required



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Development Methodology

A modular programming approach was followed to ensure maintainability and scalability

System Analysis

Understand the reservation system requirements and identify core functionalities needed

Module Division

Divide the system into 6 distinct modules, each handling specific functionality

Seat Design

Design seat structure using arrays or structures to represent 20 or 40 seats per bus

Implementation

Code each module using C++ classes, objects, and file handling for data persistence

Testing

Test booking and cancellation functionality to ensure data integrity and accuracy

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Algorithm Overview

Seat Reservation Algorithm

1. **Start**
   Begin reservation process
2. **Display Buses**
   Show available buses to user
3. **Select Bus**
   User chooses bus number
4. **Show Seats**
   Display available seats
5. **Input Seat**
   User enters seat number
6. **Check Availability**
   Verify seat is free
7. **Reserve or Error**
   Reserve seat or show already booked message
8. **Stop**
   End process

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Algorithm Overview (continued)

Ticket Cancellation Algorithm

1

Start

Begin cancellation process

2

Input Details

User enters bus number and seat number

3

Verify Booking

Check if seat is currently reserved

4

Cancel Reservation

Remove booking from system

5

Update Status

Mark seat as available

6

Stop

End process

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Project Team

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